

Survey of Information Spread Models

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Abstract

Information spread refers to the way in which information moves throughout a population. When looking at information spread on online networks specifically, this happens very quickly and can be influenced by a number of factors. The goal of modeling is to incorporate these factors so that we can create a model that gives an accurate picture of the way that information spreads online. There are multiple types of mathematical models that have been used to show this phenomenon. We survey three types of models and compare the results. The first model that is considered is an absorbing Markov chain, which is based off of the modified forest-fire model. The behavior of this matrix is analyzed through simulations and the steady-state and fundamental matrices. Next, we build a forest-fire algorithm to model the spatial aspect of information spread. We consider three versions of this algorithm, each incorporating probability in a different way, in order to see the effect on the speed of information spread. Then, the algorithm is extended to include a Burnt compartment that adds complexity to the model. For all of the algorithms, we run stochastic simulations to determine the overall trends of the model. Finally, we look at a differential equation model. It is based off of the SIR model, where information is analogous to an infectious disease. In order to better define the classes in terms of information spread, the classes will be changed to Unaware, Interested, and Disinterested. The UID model includes a reinfection rate, which allows users to move back and forth between the Interested and Disinterested classes. We perform equilibria analysis, calculate the basic reproduction number, and run simulations to determine the long-term behavior of the system. The results of these three models are compared by looking at the predicted population proportions in each class after a 24 hour time period. After this analysis, the UID model is extended to include a fourth compartment and other parameter values. This allows us to include more complex assumptions in the model. We also add a control to this model in order to attempt to maximize the number of people in the interested class.

1 Introduction

The way that information spreads on social networks is a topic that is growing in popularity today. People are very highly interconnected online and any piece of information has the ability to spread through the network at a high speed. Since there are hundreds of millions of users on any given social network, the information can easily reach a large portion of the population, even if it begins with only a small number. There are many different factors that affect how information spreads and therefore how many users receive it. For example, the number of people interested in a piece of information changes depending on the topic, so some things may spread much faster or become more widespread than others. Also, recent information is generally more interesting and engaging for the population, so that type of information spreads faster as well. By modeling how this information spread happens, we have the ability to better understand the network's users and their behavior.

One type of model that has been created to explore information spread is known as the modified forest-fire model. This model distinguishes between spreaders and non-spreaders on the network (Kumar). It will serve as a starting point for two of the models to be analyzed later. There are a large number of people that see information online and choose not to share it further, for whatever reason, so the number of users that spread the information may not actually represent the proportion of the population that is informed. The model includes four classes, where the "Fire" state represents a spreader of information and the "Burnt" state represents a non-spreader of information (Kumar). The combination of these two proportions may give a better estimate of the true proportion of the population that is informed. These four classes are adapted into an absorbing Markov chain model, where the Burnt state is the absorption state. The same forest-fire model idea is also adapted into an algorithm that visually shows the spread of information through a random network. Both the matrix model and the algorithm utilize probability to show the transition of users between states, albeit in different ways. We can compare these models to determine how predicted information spread changes when a similar idea is structured in two different ways.

Another common way that information spread is modeled is using the traditional SIR (Susceptible-Infected-Recovered) model for infectious disease spread. In this sense, the information is thought of as a "disease" that is spreading through the population. Then, the susceptible class represents a person that does not know the information, the infected class represents a person that knows about the information and is spreading it to others, and the recovered class represents a person that is no longer interested in the information. This model uses differential equations to show how the proportion of the population in each class changes over time. There are many ways that this model can be expanded in order to more accurately reflect the reality of the way information spreads online. In another model that was created to look at the spread of memes online, researchers included a reinfection rate to demonstrate how users that "recovered" could become re-interested in the information later (Wang). This modification allows for more complexity within the model. In real-world scenarios, it is often the case that there are multiple spikes in interest in a topic over a period of time, so it makes sense that the model would include becoming re-interested as an option. This model can be used to answer questions about the factors that affect information spread. For example, how many users know about the information after a 24 hour time period? In the long term, what proportion of the population is informed? Does the inclusion of a reinfection rate significantly affect either of these values as compared to other models?

(More here to talk about extending the DE model and then introducing the optimal control problem but not sure exactly what yet)

2 Markov Chain Model

2.1 Model Formulation

The first model we analyzed is an absorbing Markov chain matrix model. The states of the matrix model represent the phases of information spread that users online can move through, starting with a user that isn't on the network yet and ending with a user that has lost interest and stopped spreading the information. This type of model is unique because it is able to distinguish between individuals who spread the information and those who do not (Kumar). The population is divided into four classes: Empty, Tree, Fire, and Burnt. Empty represents a user that is not on the network at all, Tree

represents a user that is online, but does not know about the information, Fire represents a user that has the information and spreads it, and Burnt represents a user that has the information, but is no longer spreading it (Kumar). The original model uses an algorithm and calculates probabilities for each individual user, but we adapt these dynamics into a Markov Chain matrix model. There is one probability assigned to the transitions between each state that remains the same for all users. The possible transitions between states are shown below:

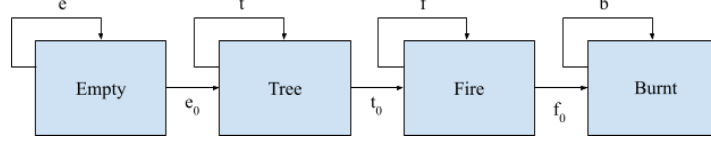


Figure 1: This compartmental diagram shows the possible transitions between states of the model.

2.2 Analysis

The matrix model can be analyzed in order to determine the long-term population proportions. These results will be compared to those of the other models later. The transition matrix can be formulated based off of the compartmental diagram and the result is shown below.

$$T = \begin{bmatrix} e & 0 & 0 & 0 \\ e_0 & t & 0 & 0 \\ 0 & t_0 & f & 0 \\ 0 & 0 & f_0 & 1 \end{bmatrix}$$

The first column represents the Empty state, the second column represents the Tree state, the third column represents the Fire state, and the fourth column represents the Burnt state. This matrix can be decomposed into the following pieces:

$$A = \begin{bmatrix} e & 0 & 0 \\ e_0 & t & 0 \\ 0 & t_0 & f \end{bmatrix}, B = \begin{bmatrix} 0 & 0 & f_0 \end{bmatrix}, O = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, I = [1]$$

Each of these matrices is important in determining the steady-state matrix, L , for the model and the fundamental matrix, F . The L matrix represents the proportion of the population in each state in the long term. The lower left corner of the matrix contains the absorption probabilities, meaning the chance that the system will be absorbed given that it started in a specific state. Since there is only one absorption state in this model, each probability in this section will end up being equal to 1. The general form of the steady-state matrix is:

$$L = \begin{bmatrix} O_{3 \times 3} & O_{3 \times 1} \\ B(I - A)^{-1} & I_{1 \times 1} \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ \frac{-e_0 f_0 t_0}{(e-1)(f-1)(t-1)} & \frac{f_0 t_0}{(f-1)(t-1)} & \frac{f_0}{1-f} & 1 \end{bmatrix}$$

Another matrix that we can analyze is the fundamental matrix. This gives information about the number of time steps that the system is in a specific state and also the average number of time steps until the system is absorbed. The general form of this matrix is:

$$F = (I - A)^{-1} = \begin{bmatrix} \frac{1}{1-e} & 0 & 0 \\ \frac{e_0}{(e-1)(t-1)} & \frac{1}{1-t} & 0 \\ \frac{-e_0 t_0}{(e-1)(f-1)(t-1)} & \frac{t_0}{(f-1)(t-1)} & \frac{1}{1-f} \end{bmatrix}$$

Each individual entry f_{ij} represents the number of time steps that the system is in state j given that it began in state i . For example, the value $\frac{1}{1-t}$ means that the system will be in the Tree state for $\frac{1}{1-t}$ time steps, given that it also started in the Tree state. Also, the sum of the entries in each column represents the average number of time steps for the system to reach the absorption state from

Parameter/Initial Condition	Value	Description
e	0.8	Probability of staying in Empty State
e_0	0.2	Probability of moving from Empty to Tree
t	0.5	Probability of staying in Tree State
t_0	0.5	Probability of moving from Tree to Fire
f	0.5	Probability of staying in Fire State
f_0	0.5	Probability of moving from Fire to Burnt
E_0	70	Initial number of users in Empty
T_0	20	Initial number of users in Tree
F_0	10	Initial number of users in Fire
B_0	0	Initial number of users in Burnt

Table 1: A description of parameters and initial conditions for the Markov chain model.

its initial state. In this case, the sum of the first column would be the average number of time steps for the system to reach the Burnt state if it initially started in the Empty state.

To make these interpretations more concrete and run simulations, we can substitute parameter values into the generalized matrices and evaluate them. The parameter values are based off of those in the modified forest-fire algorithm (Kumar). We will assume that the probabilities and initial conditions for the model are as follows:

Now, with these values, the L matrix and the F matrix become:

$$L = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 \end{bmatrix}$$

$$F = \begin{bmatrix} 5 & 0 & 0 \\ 2 & 2 & 0 \\ 2 & 2 & 2 \end{bmatrix}$$

The absorption probabilities in the L matrix mean that the Empty, Tree, and Burnt states have a 100% chance of reaching the Burnt state. Since that is the only absorption state, this makes sense in the context of the model. The individual values in the F matrix represent the average number of time steps that the model is in that state, given that it began there. For example, the value 5 means that a user will be in the Empty state for about 5 hours, given that they started in Empty. The sums of each of the columns in F are also important. They represent the number of time steps that it takes for the system to reach the absorption state given that it started in a certain state. For example, the sum of the first column is 9, which means that it will take about 9 hours for a user initially in the Empty state to reach the Burnt state. The sum of the second column is 4, which means that it will take about 4 hours for a user initially in the Tree state to reach the Burnt state. Finally, the sum of the third column is 2, which means that it will take about 2 hours for a user initially in the Fire state to reach the Burnt state.

Next, we can run simulations to see both the long term behavior of the system and the results after a 24 hour time period. Since this model includes the Burnt absorption state, in the long term, every user will end up there, but it is still interesting to look at the dynamics.

Also, we can look at the proportions of the population in each state after one day. The probability vector for the model after 24 hours is:

$$X = \begin{bmatrix} 0.0047 \\ 0.0031 \\ 0.0052 \\ 0.9869 \end{bmatrix}$$

This means that there is a 0.47% chance that the system will be in the Empty state, a 0.31% chance that the system will be in the Tree state, a 0.52% chance that the system will be in the Fire state, and a 98.69% chance that the model will be in the Burnt state. Although this model separates between

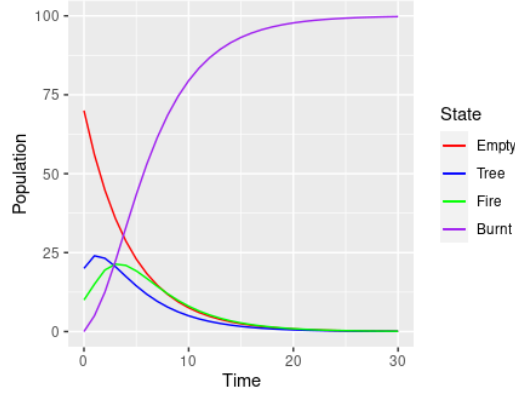


Figure 2: The long-term behavior of the Markov Chain model.

Empty and Tree states, both of these groups are unaware of the information, so in total, there is a 0.78% chance that the user will not know the information.

2.3 Limitations

Although this model demonstrates information spread through a population, there are some limitations. First, the matrix model is only based off of probability. This means that a certain proportion of users always move from one state to the next at each time step. This type of movement doesn't account for the way that people on the network are connected to each other, which would obviously impact the way that information spreads and how fast it does so. For example, if a user that is highly connected starts spreading information, it is going to reach more people than if a user with less connections starts to spread. Therefore, the time that it takes for the entire population to be informed will change depending on where the information starts from, but the matrix model does not account for this. Similarly, this model is an absorbing Markov chain, meaning that eventually all users will end up in the Burnt state. This means that all users move through all states of the model and end up in Burnt in the long term. It is not necessarily always the case that every user online starts sharing the information that they see, so the idea of every user ending up Burnt, meaning that they were sharing the information at one point, may not be accurate either.

3 Forest-Fire Algorithm

3.1 Model Formulation

In order to build off of the limitations of the Markov chain model, we will now consider a new model, the forest-fire algorithm. Since the matrix model does not account for the way that users are connected to each other during the process of information spread, this model adds a spatial aspect. The algorithm creates a network that randomly connects nodes together with a set number of edges. When a node is "informed," they can only spread the information to another node that they are directly connected to, which reflects the way that users only share content with people that follow them on social media, rather than anyone from the population.

More specifically, this algorithm relates the idea of information spread online to the way a wildfire spreads through a forest. The users are thought of as "Trees" and learning about a piece of information causes them to catch "Fire" and then spread to the people around them. We start the simulation by randomly selecting one node to start as "Fire," which starts the spread throughout the rest of the network. Since the node is randomly selected, it changes between simulations. In this model, there are only two possible states for users to be in: Tree and Fire. Since the users are interconnected on the network, once the Fire begins to spread, it will eventually reach everyone. Thus, the most interesting thing to consider is how the information spreads. We have considered three different scenarios for information spread.

First, we consider the scenario where a Fire node immediately spreads the information to all of its neighbors. We will refer to this scenario as the basic model. Since there is nothing stopping or delaying the node from sharing information, it moves through the population very quickly.

The next variation of the model introduces probability into the spread of Fire through the network. We will refer to this scenario as the node probability model. This means that the given node either spreads to all of its neighbors or it does not, based on a randomly assigned probability value. If the probability value is higher than a specified cutoff value, then the neighbors of this node will become informed, but if it is less than the cutoff value, then they will not. This value changes at each time step, so a node that does not spread it to its neighbors at one point, could spread later.

Finally, for the third variation of the algorithm, probability is incorporated once again, but in a different way. We will refer to this scenario as the neighbor probability model. A random probability value is assigned to all neighbors of a Fire node and depending on the value, that individual neighbor either becomes Fire or not. If the probability value is higher than a specified cutoff value, then that neighbor will become informed, but if it is lower, then they will remain uninformed. The probability values change at each time step, so a neighbor that doesn't receive the information initially could still learn about it at a later point. These last two scenarios incorporate the possibility of an informed person not spreading the information to everyone right away, which adds more complexity to the model.

3.2 Simulations

When running simulations of the forest-fire algorithm, the network is randomly generated each time. This means that although the number of nodes and edges remains the same each time, the way that they are connected changes. It's possible for there to be nodes that aren't connected to the rest of the network and the number of these isolated nodes can also change from simulation to simulation. Therefore, in the figures below, the shape of the network and the number of connected nodes is different. It's important to note that when we refer to the entire population being informed, we are referring to all of the nodes that are connected to one another, as there is no way for the information to ever spread to the isolated ones. These nodes can be thought of as inactive users that aren't on the network often enough to learn the information or contribute to the spread.

First, we will look at the graphs of the basic model. In this model, the information automatically spreads from an informed node to all of its neighbors, so the entire population is informed fairly quickly. For this particular simulation, it takes 7 time steps, which are shown below.

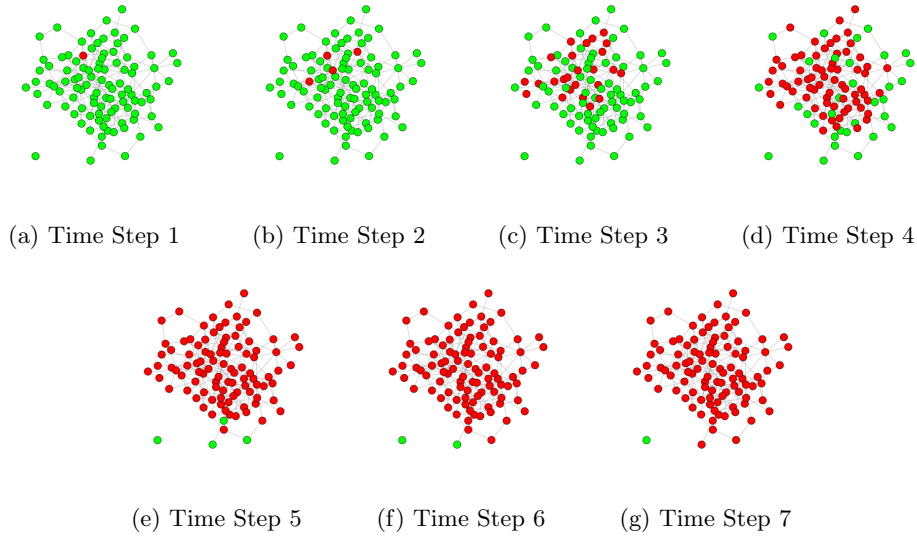


Figure 3: The network graphs of one simulation of the basic model.

Next, we consider the network graphs of the node probability model. This is the first version of the algorithm that incorporates probability, which will slow the spread of information. Thus, in this

simulation, it takes 16 time steps for the entire population to be informed.

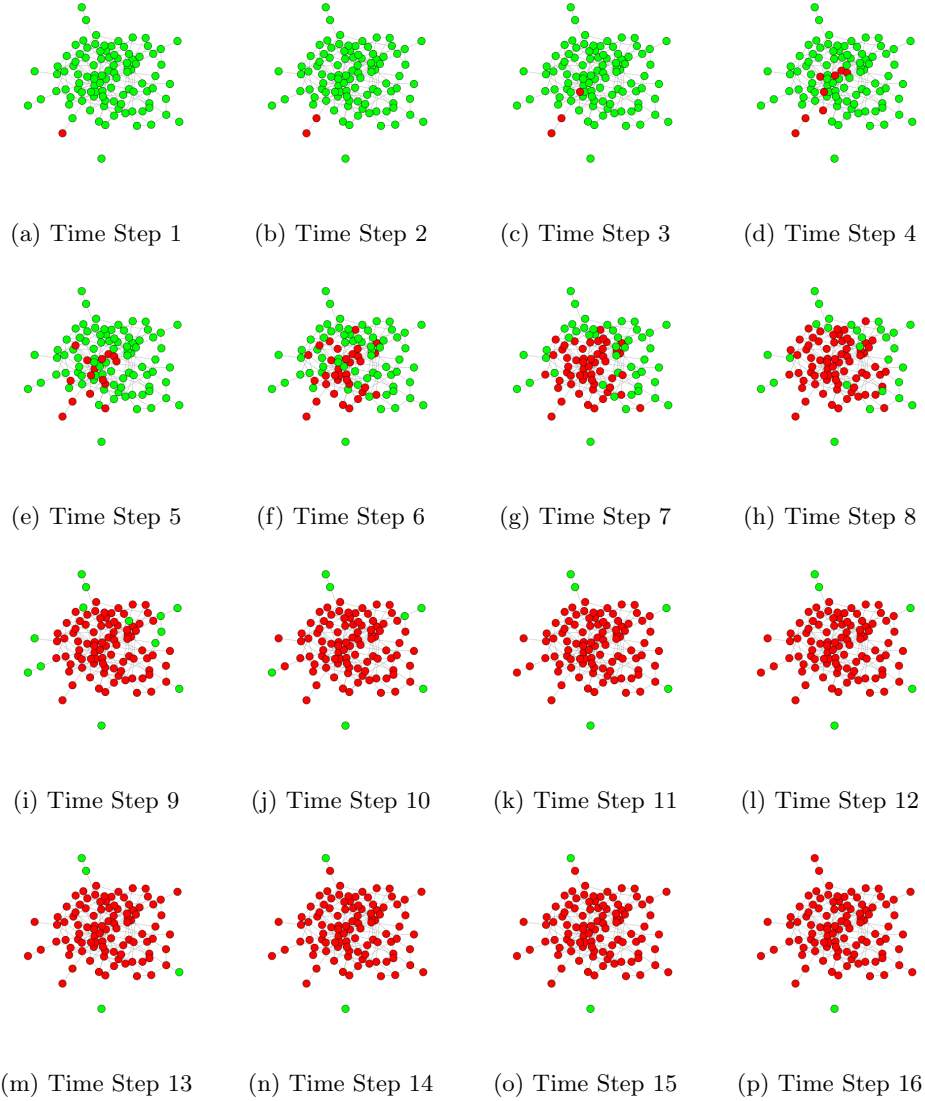


Figure 4: The network graphs of one simulation of the node probability model.

Finally, we look at the networks graphs of the neighbor probability model. This algorithm incorporates probability in a different way, but it still slows the spread of information through the population. It takes 12 time steps for the information to reach the entire population, which is longer than in the basic model, but shorter than the node probability model.

3.3 Analysis

In order to further analyze this model, we also ran stochastic simulations. Since the node that initially starts as Fire is randomly selected, it could be a node that is highly connected to others, or one that is not. It makes sense that a node with lots of neighbors will spread the information through the population much faster than one with only a few neighbors. Thus, we run many stochastic simulations in order to see the overall trends of each version of the algorithm, rather than just one case.

The first variation of this model is the most basic because it does not incorporate any probability into the spread of Fire through the network. In this case, every neighbor of a node that is on Fire automatically becomes Fire as well. Since most nodes have multiple neighbors, this means that Fire spreads through the network very quickly. Although this is generally true, it is important to note that

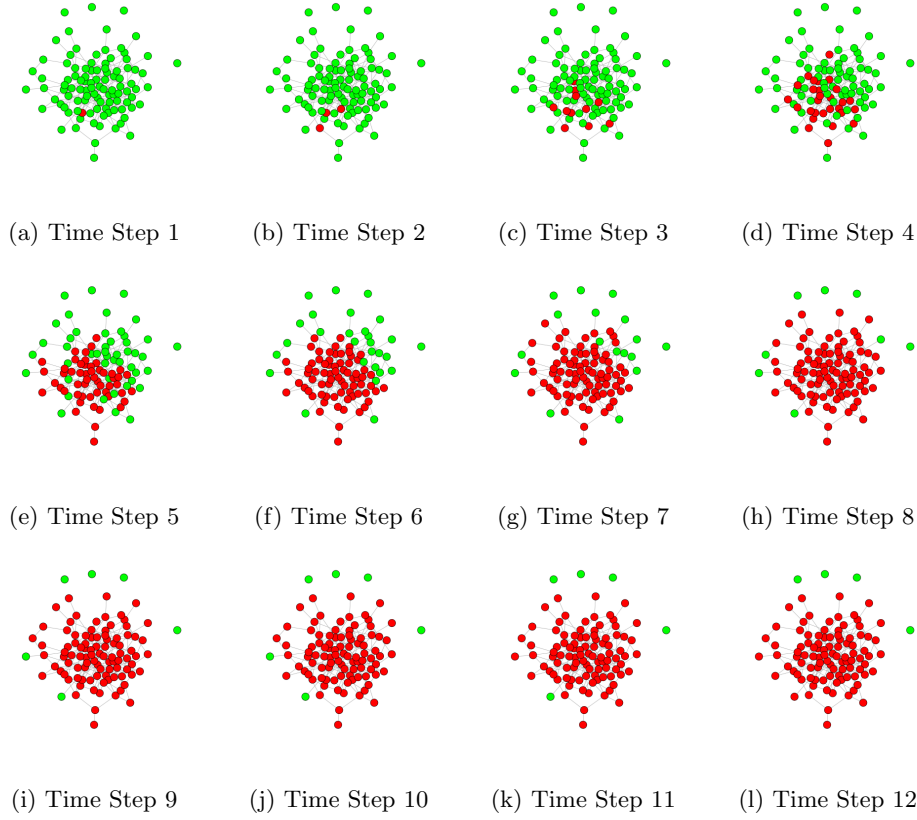


Figure 5: The network graphs of one simulation of the neighbor probability model.

the initial Fire node is randomly selected, which could still impact the spread. In order to look at overall trends of the model, 100 simulations were done and the results are shown in the figure below.

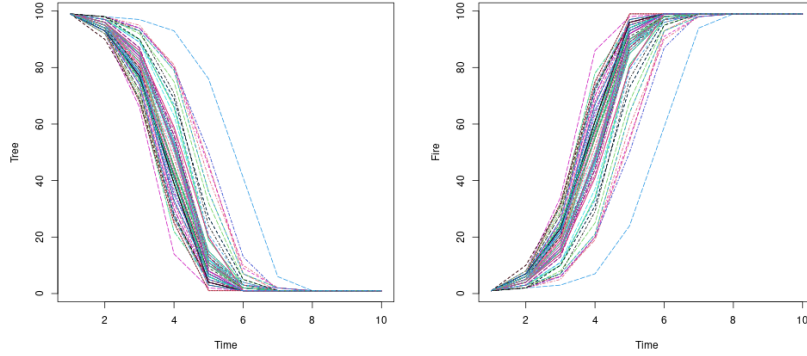


Figure 6: The results of 100 simulations of the basic model.

These graphs are a way to visually show the variation that occurs from the stochastic simulations. Every simulation begins and ends with the same number of nodes categorized as Tree or Fire, but depending on which node was selected initially, the way that the information spreads throughout the population is slightly different each time. In order to more clearly see the overall trends of the simulations, we can take the average of all of them. The results are shown in both the figure and table below.

These figures allow us to make some conclusions about the spread of information through this version of the algorithm. Although there is variation in the different simulations, in each one, the

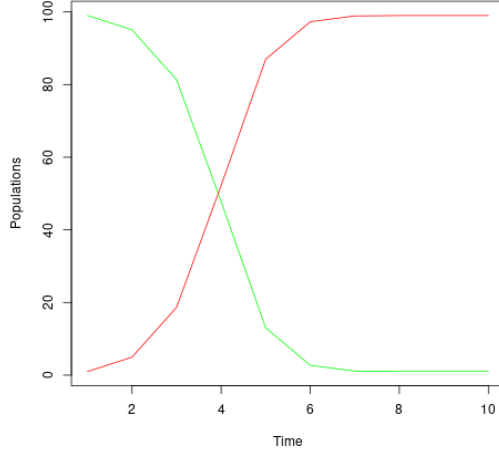


Figure 7: The average result of 100 simulations of the basic model.

Time Step	Tree	Fire
1	99.00	1.00
2	95.05	4.95
3	81.33	18.67
4	47.79	52.21
5	13.04	86.96
6	2.70	97.30
7	1.11	98.89
8	1.00	99.00
9	1.00	99.00
10	1.00	99.00

Table 2: The average number of Tree and Fire at each time step of the basic model.

entire population was informed by time step 8. It's likely that this happened faster in some scenarios, but these averages tell us that it took no longer than 8 time steps in every iteration of the simulation. In the context of an social network, this means that everyone online has heard about the information within 8 hours, if not sooner. More analysis was done to find a precise estimate for the average amount of time it takes for the entire population to be informed. For each simulation, we recorded the time step when all connected nodes have become Fire and then took the average of all of them. This tells us that it takes about 6.52 time steps, on average, for the entire population to be informed using the basic model.

The next variation of the model is the node probability model. It allows for an informed node to either spread to its neighbors or not, based on a probability value. Since there is the chance that the Fire may or may not spread at different times, it ends up taking longer to reach the entire population. Also, the introduction of probabilities means that there is more variability in the model. In some cases, the probabilities may all be very high, meaning that the information will spread quickly, but in other cases, the probabilities may be lower, meaning that the spread will take longer. Again, 100 simulations were done and the results are shown below.

There is clearly a wider range of values that was produced in these simulations than in the prior model. This is again due to the randomly selected initial Fire node and also the random probability of a node spreading. The spread also happened slower than in the previous model. The general shape of the graphs is similar, but the time scale is different. We are looking at the results of the simulations over 20 time steps, which is twice as long as the basic model. Since there is again variation in the simulations, it is helpful to look at the averages to help us summarize the results. These are shown in the figure and table below.

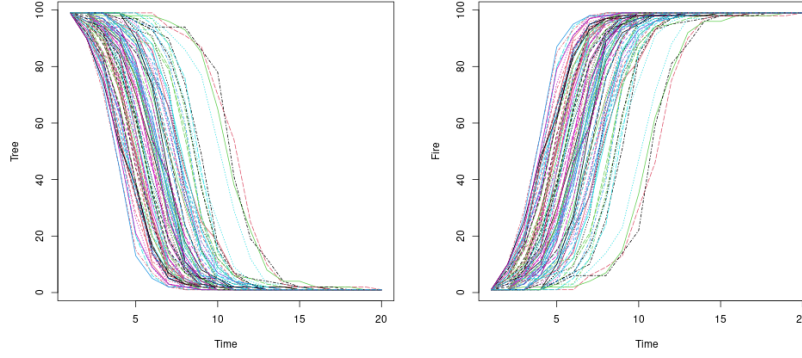


Figure 8: The results of 100 simulations of the node probability model.

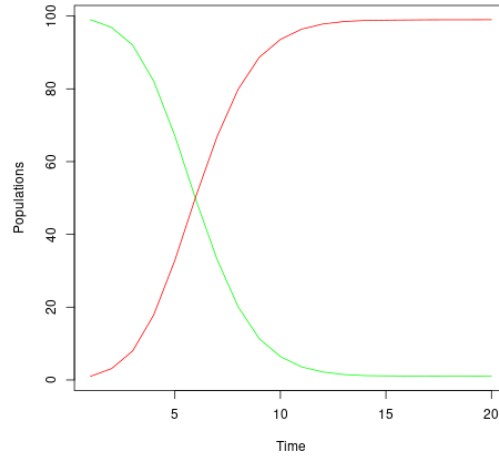


Figure 9: The average result of 100 simulations of the node probability model.

It's clear from the figures that the information spread happens more slowly in this version of the algorithm, but with the averages, we can tell more specifically how different the results are. The table tells us that it is not until time step 20 that every simulation shows the entire population as being informed. Obviously not every simulation happens this slowly, but it is over twice as long as the estimate of 8 from the previous model. Again, we can find a more precise answer by recording the time steps required in each simulation and taking the average of them. This gives us a result of 11.95 time steps until the entire population is informed, which is over 5 time steps longer than the average time from the basic model. Thus, it's clear that the addition of probability into the model slows the spread of information through the population. Also, since there is a fairly large difference between the average number of time steps until the population is informed and the number of time steps until the population is informed in every simulation, this tells us that there is significantly more variation in the stochastic simulations for the node probability model.

Finally, we considered the neighbor probability model. This model also incorporates probability, but in a different way than in the node probability model. For a node that is on Fire, we assign a random number between zero and one to each individual neighbor. If that value is greater than 0.5, that specific neighbor turns to Fire, but otherwise it remains a Tree. This process is repeated for all neighbors of the initial fire node. This means that the fire may spread to some neighbors and not others, which would slow down the spread throughout the population. The probability values reset at each time step, so a neighbor that doesn't learn the information at one time step is able to become informed at a later point. This is a similar concept to that of the node probability model, except that the changing probability values are assigned to the neighbors of the Fire node, rather than the

Time Step	Tree	Fire
1	99.00	1.00
2	96.87	3.13
3	92.02	7.98
4	82.10	17.90
5	67.11	32.89
6	49.46	50.54
7	33.13	66.87
8	20.09	79.91
9	11.33	88.67
10	6.42	93.58
11	3.60	96.40
12	2.20	97.80
13	1.50	98.50
14	1.21	98.79
15	1.13	98.87
16	1.08	98.92
17	1.03	98.97
18	1.02	98.98
19	1.01	98.99
20	1.00	99.00

Table 3: The average number of Tree and Fire at each time step of the node probability model.

Fire node itself. We are interested in seeing how this slight change impacts the spread of information through the network. There were 100 simulations run for this model and the results are shown below.

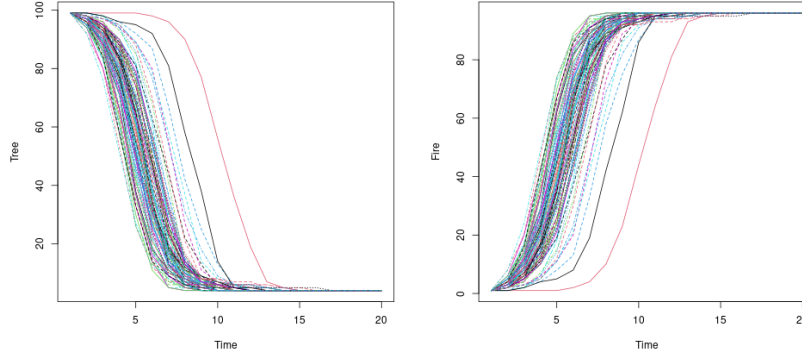


Figure 10: The results of 100 simulations of the neighbor probability model.

Given the similarities between this model and the node probability model, it makes sense that the results are fairly similar as well. The time frame we are looking at is 20 time steps, which is the same as the node probability model, but the entire population appears to become informed slightly earlier than in that model. The table confirms this thought. It tells us that the entire population was informed by time step 17 in every simulation, although this likely happened faster in some cases. After doing further analysis, we find the average number of time steps until the population is informed through this model is 11.22. This value is much closer to the average number of time steps for the node probability model, which was 11.95. It's interesting that these values are so close to one another, given the changes between the two models. However, it does make sense that between the two, the information takes longer to spread in the node probability model. In this case, the information either spreads from one informed user to all of their neighbors or to none at all, whereas in the neighbor probability model, the information can spread to some number in the middle. This option could mean that the information reaches some other users faster than in the previous model, but the averages tell

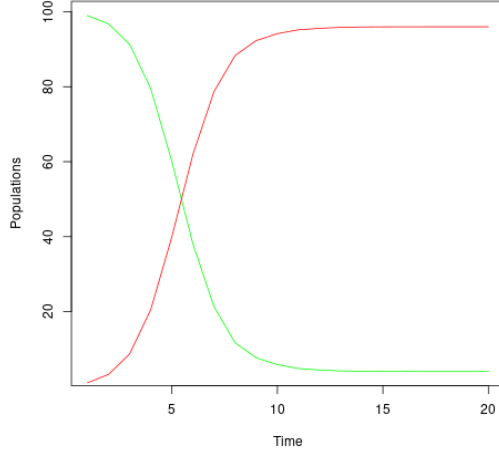


Figure 11: The average result of 100 simulations of the neighbor probability model.

Time Step	Tree	Fire
1	99.00	1.00
2	96.77	3.23
3	91.29	8.71
4	79.47	20.53
5	60.04	39.96
6	38.00	62.00
7	21.25	78.75
8	11.63	88.37
9	7.65	92.35
10	5.82	94.18
11	4.80	95.20
12	4.41	95.59
13	4.15	95.85
14	4.05	95.95
15	4.02	95.98
16	4.01	95.99
17	4.00	96.00
18	4.00	96.00
19	4.00	96.00
20	4.00	96.00

Table 4: The average number of Tree and Fire at each time step of the neighbor probability model.

us that it doesn't create a significant difference.

3.4 Burnt Model Extension

3.4.1 Model Formulation

The forest-fire algorithm has been extended to include a third compartment that users can move into. We call this the "Burnt" state, referring to a user that learns about the information, but chooses not to spread it any further. Now, when there is an informed node, its neighbors that are still Trees have the option of either turning to Fire and continuing the spread of information, or becoming Burnt. Whether a node becomes Fire or Burnt is based off of a randomly assigned probability value. Once a node becomes Burnt, it remains that way, so there is no further spread of information from that user. This means that the neighbors of this node will not receive the information, unless they happen

to be connected elsewhere. Unlike the previous versions of the algorithm, a node that learns about the information does not necessarily have to spread it to the rest of the population. Since there is the option to either continue the spread or stop, the information may not reach the entire population in this scenario. It is possible that enough users will lose interest and the information spread will die out before everyone has heard about it.

3.4.2 Simulations

Similarly to the previous iterations of the forest-fire algorithm, we can look at the graphs of the network to see the progression of the information spread throughout the population. However, in this case, there is not a point where the entire population is informed, because of the addition of the Burnt state, so we look at each time step until the proportion of the population in each state is no longer changing. Depending on which node initially begins as Fire and the random probability values, this will change between simulations, but one example is shown below.

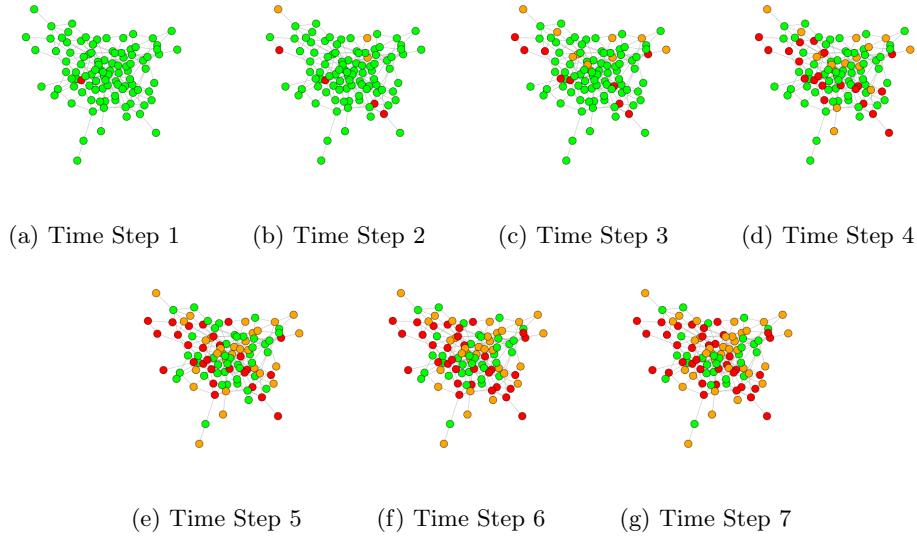


Figure 12: The network graphs of one simulation of the burnt model extension.

3.4.3 Analysis

Again, we ran stochastic simulations to look at the overall trends of the model. However, since the burnt extension adds a third state to the model, there are several differences between these simulations and ones from the previous versions of the model. We still start out with one randomly selected node on fire, but the neighbors of this node can either become Fire or Burnt, not just Fire like in previous models. This means that depending on the random probability values that are generated, the information can continue to spread to most of the population, or it could die out very quickly. This creates a lot more variation in the final number of nodes in each state. The addition of the Burnt state means that there is not a single state that the entire population is moving into, so the ending proportions are different every time. The results of 100 simulations of the burnt extension model are shown below.

Although there is much more variation in this extension of the model, we can still determine some general trends. For example, we always start with the highest number of Trees, which then decreases as time goes on. We start with one Fire node and this number increases as the information spreads throughout the population. Finally, there are not any Burnt nodes to begin with, but we begin to see more as the number of Fire nodes increases and some of the users exposed to the information are uninterested. These behaviors happen for most of the simulations, but the figure shows that there are obvious outliers. Thus, we take the average of all 100 simulations to have a clearer picture of what is happening in the model. The results are shown in the figure and table below.

With these averages, we can tell that we end up with the fewest number of Trees and the highest number of Fire, with the number of Burnt users being in the middle. However, all of these values

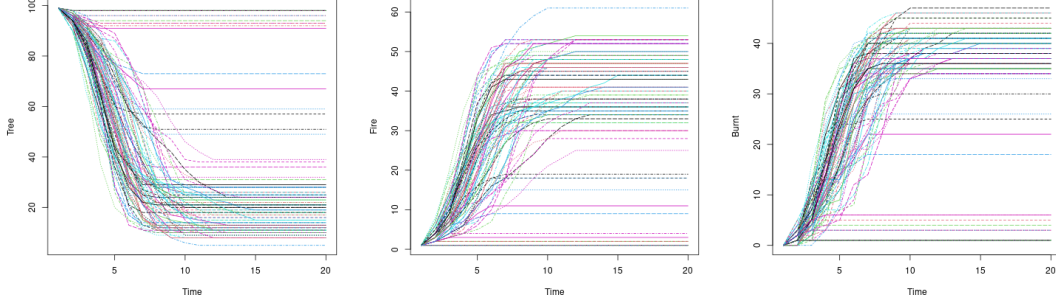


Figure 13: The results of 100 simulations of the burnt extension model.

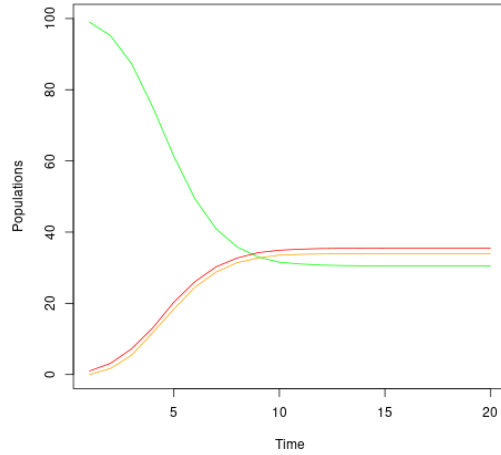


Figure 14: The average result of 100 simulations of the burnt extension model.

are close together and each state represents around one third of the population. This means that this model predicts about one third of the population will be actively spreading the information online, one third of the population will hear about the information but not spread it, and one third won't know about the information at all. These proportions may or may not be entirely realistic, but it is fair to say that they more accurately reflect real-life scenarios than the previous versions of the model. It's unlikely that every user that comes into contact with the information is going to spread it to everyone they know and that this spread is going to continue constantly until every user on the network knows about it. Thus, the burnt extension model adds complexity and more closely reflects what we would expect to see in real-life scenarios.

3.5 Limitations

Although the forest-fire algorithm and the burnt extension builds off of the limitations of the Markov chain model, there are still some scenarios that it cannot incorporate. There is the spatial component of users being connected on a network and the burnt extension allows users to choose not to spread the information. However, the users don't have a chance to move between states once they encounter the information. Depending on the probability value, they either become Fire or Burnt and then they stay in that compartment for the remainder of the simulation. On a social network, it's fair to assume that a user may start spreading the information, but then lose interest and stop shortly thereafter. Or, they may not spread the information initially, but then something may cause them to become re-interested and then they will spread it later. The algorithms we've created don't account for these situations. Thus, the following model incorporates a reinfection rate that allows users to move between states of information spread.

Time Step	Tree	Fire	Burnt
1	99.00	1.00	0.00
2	95.16	3.15	1.69
3	87.28	7.25	5.47
4	75.10	13.10	11.80
5	61.30	20.32	18.38
6	49.36	26.10	24.54
7	40.93	30.26	28.81
8	35.83	32.74	31.43
9	33.00	34.25	32.75
10	31.54	34.92	33.54
11	31.04	35.21	33.75
12	30.73	35.39	33.88
13	30.61	35.46	33.93
14	30.55	35.48	33.97
15	30.52	35.50	33.98
16	30.52	35.50	33.98
17	30.52	35.50	33.98
18	30.52	35.50	33.98
19	30.52	35.50	33.98
20	30.52	35.50	33.98

Table 5: The average number of Tree and Fire at each time step of the burnt extension model.

4 Basic ODE Model

4.1 Model Formulation

The classic Susceptible-Infected-Recovered (SIR) model divides the population into three groups. While it is usually used in terms of infectious disease spread, these concept can also be applied to information spread online. In order to better fit the information spread context, titles of the classes will be changed to become Unaware, Interested, and Disinterested. The unaware class refers to individuals that do not know the information. The interested class refers to individuals that know the information and are actively spreading it to others. The disinterested class refers to individuals that are no longer spreading the information. The equations for this model are as follows:

$$\frac{dU}{dt} = -\beta_I UI \quad (1)$$

$$\frac{dI}{dt} = \beta_I UI - \alpha I + \gamma D \quad (2)$$

$$\frac{dD}{dt} = \alpha I - \gamma D \quad (3)$$

In order for users to move from unaware to interested, they must have an interaction with an already interested user. These interactions occur at the rate β_I , which is the transmission coefficient. Once informed users lose interest and stop spreading the information, they move to the disinterested class at a recovery rate α (Smith). This model assumes that the parameters α and β_I do not change, so the proportion of the population that becomes interested or disinterested stays the same at each time step. Next, it is possible for users to regain interest and return to the interested class at a later point, which occurs at a reinfection rate γ (Wang).

4.2 Equilibria Analysis

One way we can analyze the behavior of this model is by looking at the equilibria. First, in order to simplify our analysis, we assume that the population size, N , is constant. This allows us to write the third compartment of the system as $D = N - U - I$ and reduce the system to two equations. The simplified model is shown below:

$$\frac{dU}{dt} = -\beta_I UI \quad (4)$$

$$\frac{dI}{dt} = \beta_I UI - \alpha I + \gamma(N - U - I) \quad (5)$$

From here, we find the equilibria of the model to be $(N, 0)$ and $(0, \frac{\gamma N}{\alpha + \gamma})$. The second equilibrium is the endemic one, meaning the case where the information is continuing to spread through the population, so this is where we will focus our analysis. In order to determine the stability of an equilibrium, we must calculate for the Jacobian for the system.

$$J = \begin{bmatrix} -\beta_I I & -\beta_I U \\ \beta_I I - \gamma & \beta_I U - \alpha - \gamma \end{bmatrix}$$

This matrix can be evaluated for the equilibria and the eigenvalues can be used in order to determine stability. The Jacobian for the equilibrium $(0, \frac{\gamma N}{\alpha + \gamma})$ is:

$$J(0, \frac{\gamma N}{\alpha + \gamma}) = \begin{bmatrix} \frac{-\beta_I \gamma N}{\alpha + \gamma} & 0 \\ \frac{\beta_I \gamma N}{\alpha + \gamma} - \gamma & -\alpha - \gamma \end{bmatrix}$$

Since this matrix is triangular, the eigenvalues are found to be $\frac{-\beta_I \gamma N}{\alpha + \gamma}$ and $-\alpha - \gamma$. Since the parameter values in this model are positive, both of these eigenvalues will be negative. This means that the equilibrium is a sink, which is stable.

4.3 Basic Reproduction Number

When used in terms of infectious diseases, the basic reproduction number, denoted R_0 , represents the contagiousness of a disease by predicting the number of secondary cases that will come from a single infection (Delamater). This value can still be a useful statistic in the context of information spread. We can think of a piece of information spreading through the population like an infectious disease and an "infection" is a new user that learns about the information and in turn spreads it to other people. In general, an R_0 greater than 1 means that the disease, or piece of information, will continue to spread throughout the population, and an R_0 less than 1 means that it will eventually die out.

We can calculate this value based on our model. The equations of the system are separated into matrices $\mathcal{F}$ and $\mathcal{V}$. The matrix $\mathcal{F}$ includes terms that are related to new infections, while all other terms are placed in $\mathcal{V}$. The matrices can be written as

$$\mathcal{F} - \mathcal{V} = \begin{bmatrix} \beta_I UI \\ 0 \\ 0 \end{bmatrix} - \begin{bmatrix} \alpha I - \gamma D \\ \beta_I UI \\ -\alpha I + \gamma D \end{bmatrix}.$$

Then, we take the Jacobian of the $\mathcal{F}$ and $\mathcal{V}$ matrices. Since interaction with the I compartment is what leads to new users becoming informed, the partial derivative will be taken with respect to I. Once these matrices are evaluated at the disease free equilibrium, they become:

$$F = [\beta_I N] \quad \text{and} \quad V = [\alpha].$$

Next, we will calculate V^{-1} :

$$V^{-1} = [\frac{1}{\alpha}].$$

Finally, we calculate FV^{-1} in order to find R_0 .

$$FV^{-1} = [\beta_I N] [\frac{1}{\alpha}] = [\frac{\beta_I N}{\alpha}]$$

The R_0 value is defined as the spectral radius of FV^{-1} , which is the dominant eigenvalue of the matrix. Since there is only one element in this matrix, we can easily determine R_0 to be:

$$R_0 = \frac{\beta_I N}{\alpha}$$

With a given set of parameter values, we can calculate the numerical value for R_0 . They will be discussed in more detail in the next section, but let $N = 100$, $\beta_I = 0.005$, $\alpha = 0.1$, and $\gamma = 0.05$. After plugging in, we get that $R_0 = 5$. Since this value is greater than 1, we see that the information will continue to spread throughout the population.

4.4 Simulations

Another way to analyze this model is to run simulations. We will assume a population size of $N = 100$ and parameter values of $\beta_I = 0.005$, $\alpha = 0.1$, and $\gamma = 0.05$. These values were chosen as test values to create dynamics similar to those one might see in a real information spread scenario. The transmission of the information happens quickly and the system ends up stabilizing at an equilibrium value. The dynamics are shown in the figure below.

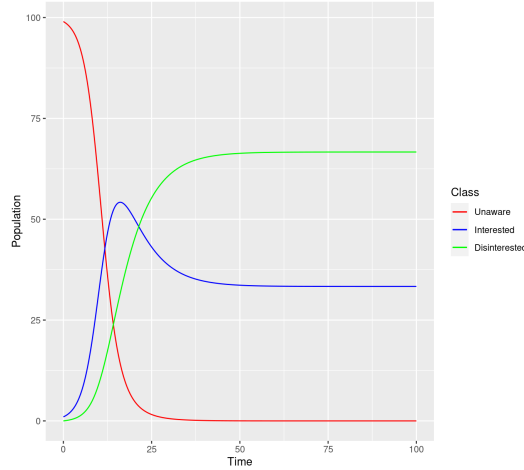


Figure 15: The long-term behavior of the differential equation model.

We see that the peak of the interested class happens before 24 hours and includes just over half of the population. Then, as time goes on, the number of people talking about the information decreases and eventually levels out at one-third of the population. The number of users in the disinterested class steadily increases over time and levels out with two-thirds of the population. This means that there is no one left in the unaware class. Going back to the endemic equilibrium we found earlier, the value $(0, \frac{\gamma N}{\alpha + \gamma})$ becomes $(0, 33\frac{1}{3})$ after plugging in parameter values, which matches what we see in our dynamics. Since we made the substitution $D = N - U - I$, we can now see that $D = 66\frac{2}{3}$ in the long term. When thinking about the context of information spread, these values make sense. A new piece of information will be interesting and exciting for users, so many people will be talking about it and spreading it to others. However, as the novelty wears off, users will slowly stop spreading the information, or choose to not spread it at all. Thus, we see that in the long term there is twice as many users in the disinterested class as there are in the interested class.

5 Model Comparison

Although the three models discussed above are all different, there are still some comparisons that can be made between them. One way to do this is by looking at the predicted proportions of the population in each compartment after 24 hours. Starting with the Markov chain model, we can see there is a 0.41% chance of being in the Empty state, a 0.31% chance of being in the Tree state, a 0.52% chance of being in the Fire state, and 98.69% chance of being in the Burnt state. In order to better compare with the other two models, we will look at the burnt extension version of the forest-fire algorithm because it includes a compartment for users that aren't sharing the information. After taking the averages between all of the stochastic simulations, we see that the proportions level out with 30.52% of the population in the Tree compartment, 35.50% of the population in the Fire compartment, and 33.98% of the population in the Burnt compartment. These values stabilize before we reach 24 hours,

but we can assume that they will continue to stay the same. Finally, in the UID model, we end up with 1.97% of the population in the unaware class, 44.37% of the population in the interested class, and 53.65% of the population in the disinterested class.

There are clear differences in the values between the three models, but given the different structures of the models, they make sense. For example, the Markov chain includes the Burnt state as an absorption state, meaning that all users will end up there eventually and once a user loses interest in the information, they stay disinterested forever. Therefore, it makes sense that the proportion of the population in this class is much higher than in the other two models. In the burnt extension algorithm and the UID model, the burnt and disinterested compartments are designed differently. In the algorithm, once a user becomes Burnt, they stay that way, but it is not an absorption state, so users aren't all moving towards that compartment like in the matrix model. For the UID model, users are able to enter and leave the disinterested compartment over time as they lose or regain interest. There are still some differences between these two models, but the lack of an absorption state makes them more comparable. Thus, it makes sense that the proportions after 24 hours are much closer together, with 33.98% of the population in the burnt state of the forest-fire algorithm and 53.65% of the population in the disinterested class of the UID model.

6 Extended ODE Model

Then, the model is expanded to include a fourth compartment. This class, W , represents a hub of information on the Internet. Informed users can contribute to this hub and unaware users can now become informed either through a direct interaction with an informed user or through contact with the hub. (Probably need more here about what W means- Water in a biological context? And how that same idea can be applied to information spread)

$$\frac{dU}{dt} = -\beta_I UI - \beta_W UW \quad (6)$$

$$\frac{dI}{dt} = \beta_I UI + \beta_W UW - \alpha I + \gamma D \quad (7)$$

$$\frac{dD}{dt} = \alpha I - \gamma D \quad (8)$$

$$\frac{dW}{dt} = \sigma I - \delta W \quad (9)$$

There are now two terms creating new infections, $\beta_I UI$ and $\beta_W UW$, because there are two ways for users to learn about the information. The parameters α and γ are the same as in the previous model. The information hub, W , grows through contributions from informed users, but there is also a decay of interest in the information once there is too much of it, represented by the parameter δ .

This model assumes that all users that learn about the information move to the Informed class and continue spreading the it. However, this may not always be the case on a social network. It is possible for users to learn about the information, but not be interested enough to share it with others. So, we adapt the model to allow users that learn the information to move straight to the Disinterested class.

$$\frac{dU}{dt} = -\beta_I UI - \beta_W UW \quad (10)$$

$$\frac{dI}{dt} = s\beta_I UI + t\beta_W UW - \alpha I + \gamma D \quad (11)$$

$$\frac{dD}{dt} = (1-s)\beta_I UI + (1-t)\beta_W UW + \alpha I - \gamma D \quad (12)$$

$$\frac{dW}{dt} = \sigma I - \delta W \quad (13)$$

The values s and t represent the proportions of the population that move into the Informed class after learning the information from either another user or the hub, while the values $(1-s)$ and $(1-t)$ represent the proportions of the population that move into the Disinterested class.

Next, we consider the possibility of users becoming overwhelmed by too much information online. Once this occurs, they are likely to ignore it and no longer spread. To incorporate this into the model, we use the Hill function. This type of function allows for a saturation effect, meaning that the transmission of information is higher when there is less information on the hub and lower when there is more information on the hub.

$$\frac{dU}{dt} = -\beta_I UI - \frac{\beta_W UW}{M+W} \quad (14)$$

$$\frac{dI}{dt} = \beta_I UI + \frac{\beta_W UW}{M+W} - \alpha I + \gamma D \quad (15)$$

$$\frac{dD}{dt} = \alpha I - \gamma D \quad (16)$$

$$\frac{dW}{dt} = \sigma I - \delta W \quad (17)$$

Now, we will combine the extensions from the previous two models. After users learn about the information, they can either move to Informed or straight to Disinterested, and there is also a saturation effect of having too much information on the hub. Both of these extensions model scenarios that affect the proportion of users that are actively spreading information, so by combining them, we create a model that considers multiple possibilities at once and therefore gives a more accurate picture of real-world dynamics.

$$\frac{dU}{dt} = -\beta_I UI - \frac{\beta_W UW}{M+W} \quad (18)$$

$$\frac{dI}{dt} = s\beta_I UI + \frac{t\beta_W UW}{M+W} - \alpha I + \gamma D \quad (19)$$

$$\frac{dD}{dt} = (1-s)\beta_I UI + \frac{(1-t)\beta_W UW}{M+W} + \alpha I - \gamma D \quad (20)$$

$$\frac{dW}{dt} = \sigma I - \delta W \quad (21)$$

This model can be extended even further to incorporate other assumptions that reflect reality. Some of these additions include an influx of users into the population, a different type of reinfection rate, and a complete loss of interest rate. The updated model is shown below.

$$\frac{dU}{dt} = \Delta - \beta_I UI - \frac{\beta_W UW}{M+W} \quad (22)$$

$$\frac{dI}{dt} = s\beta_I UI + \frac{t\beta_W UW}{M+W} - \alpha I + \gamma ID \quad (23)$$

$$\frac{dD}{dt} = (1-s)\beta_I UI + \frac{(1-t)\beta_W UW}{M+W} + \alpha I - \gamma ID - \mu D \quad (24)$$

$$\frac{dW}{dt} = \sigma I - \delta W \quad (25)$$

The first extension of the model is the influx of users into the population, Δ . Online networks are constantly expanding, so it makes sense that the population size would not remain constant. Since these users are just joining the network, we assume that they are uninformed. They are placed into the Unaware class to start, and can then move to Informed or Disinterested like the rest of the population. Next, we modified the reinfection rate to be γID , rather than just γD . This means that in order for a user to become re-interested in the information, they must have a direct interaction with an Informed person. In the previous models, they were able to become re-interested on their own. This extension makes sense because it's likely the case that a user would have to interact with an informed person in some way, like receiving a message or seeing a post, in order to start talking about the information again, instead of just deciding by themselves to start spreading. Finally, we added a removal rate, μD to the model. In this case, the users are being removed from the possibility of reinfecting. In previous versions of the model, users stayed in either Informed or Disinterested and continued to move back

Parameters	Description
β_I	transmission rate of informed individual
β_W	transmission rate of information hub
α	loss of interest rate
γ	reinfection rate
σ	contribution rate of informed individual to information hub
δ	decay of information hub
M	half-saturation constant
s	proportion of population that spreads information from individual contact
t	proportion of population that spreads information from contact with hub
μ	removal rate
Δ	influx of new users into the population

Table 6: Parameter description

and forth between the two in the long term. However, it's reasonable to assume that eventually users will be completely disinterested in the information and will never return to sharing it because they have already seen it so many times. Thus, these users are removed at rate μ and there are fewer users able to become reinfected.

6.1 Equilibria Analysis

Do these change based off of the newest model? Need to check What equilibria are and maybe stability?

DFE: $(N, 0, 0, 0)$

Endemic Equilibrium: $(0, \frac{\gamma(I_0+D_0)}{\gamma+\alpha}, \frac{\alpha(I_0+D_0)}{\gamma+\alpha}, \frac{\gamma\sigma(I_0+D_0)}{\delta(\gamma+\alpha)})$

6.2 Basic Reproduction Number

Needs to be updated for newest model

Talk about what R_0 is in general.

The equations are separated into matrices $\mathcal{F}$ and $\mathcal{V}$. The matrix $\mathcal{F}$ includes terms that are related to new infections, while all other terms are placed in $\mathcal{V}$. The matrices can be written as

$$\mathcal{F} - \mathcal{V} = \begin{bmatrix} s\beta_I UI + \frac{t\beta_W UW}{M+W} \\ 0 \\ 0 \\ 0 \end{bmatrix} - \begin{bmatrix} \alpha I - \gamma D \\ -\sigma I + \delta W \\ \beta_I UI + \frac{\beta_W UW}{M+W} \\ -(1-s)\beta_I UI - \frac{(1-t)\beta_W UW}{M+W} - \alpha I + \gamma D \end{bmatrix}.$$

Then, we take the Jacobian of the $\mathcal{F}$ and $\mathcal{V}$ matrices. Since interaction with the I and W compartments is what leads to new users becoming informed, the partial derivative will be taken with respect to I and W. Once these matrices are evaluated at the disease free equilibrium, they become:

$$F = \begin{bmatrix} s\beta_I N & \frac{t\beta_W N}{M} \\ 0 & 0 \end{bmatrix} \quad \text{and} \quad V = \begin{bmatrix} \alpha & 0 \\ -\sigma & \delta \end{bmatrix}.$$

Next, we will calculate V^{-1} :

$$V^{-1} = \frac{1}{\alpha\delta} \begin{bmatrix} \delta & 0 \\ \sigma & \alpha \end{bmatrix} = \begin{bmatrix} \frac{1}{\alpha} & 0 \\ \frac{\sigma}{\alpha\delta} & \frac{1}{\delta} \end{bmatrix}.$$

Finally, we calculate FV^{-1} in order to find R_0 .

$$FV^{-1} = \begin{bmatrix} s\beta_I N & \frac{t\beta_W N}{M} \\ 0 & 0 \end{bmatrix} \begin{bmatrix} \frac{1}{\alpha} & 0 \\ \frac{\sigma}{\alpha\delta} & \frac{1}{\delta} \end{bmatrix} = \begin{bmatrix} \frac{s\beta_I N}{\alpha} + \frac{t\beta_W \sigma N}{\alpha\delta M} & \frac{t\beta_W N}{\delta M} \\ 0 & 0 \end{bmatrix}$$

The R_0 value is defined as the spectral radius of FV^{-1} , which is the dominant eigenvalue of the matrix. Thus, we determine R_0 to be:

$$R_0 = \frac{N(s\beta_I\delta M + t\beta_W\sigma)}{\alpha\delta M}$$

Also here are the R_0 values for the previous models, just to compare

$$Model1 : R_0 = \frac{\beta_I N}{\alpha}$$

$$Model2 : R_0 = \frac{N(\beta_I\delta + \beta_W\sigma)}{\alpha\delta}$$

$$Model3 : R_0 = \frac{N(s\beta_I\delta + t\beta_W\sigma)}{\alpha\delta}$$

$$Model4 : R_0 = \frac{N(\beta_I\delta M + \beta_W\sigma)}{\alpha\delta M}$$

6.3 Simulations

Come back once we have set parameter values

7 Optimal Control Problem

7.1 Model Formulation

Looking to maximize I . Our control is $k(t)$, which we add to the information hub, W , in order to inform more members of the population. The objective functional is:

$$\max_k J(k) = \max_k \int_0^T c_0 I - c_1 k - c_2 k^2 dt \quad (26)$$

The Hamiltonian is formed by adding the right hand side of the state equations to the objective functional using an adjoint (λ_i) .

$$\begin{aligned} H = & c_0 I - c_1 k - c_2 k^2 \\ & + \lambda_1 \left[-\beta_I U I - \frac{\beta_W U W}{M + W} \right] \\ & + \lambda_2 \left[s\beta_I U I + \frac{t\beta_W U W}{M + W} - \alpha I + \gamma D \right] \\ & + \lambda_3 \left[(1 - s)\beta_I U I + \frac{(1 - t)\beta_W U W}{M + W} + \alpha I - \gamma D \right] \\ & + \lambda_4 [\sigma I - \delta W + k] \end{aligned} \quad (27)$$

Next we take partial derivatives to form the adjoint equations.

$$\lambda'_1 = -\frac{dH}{dU} = \lambda_1 \left[\beta_I I + \frac{\beta_W W}{M + W} \right] - \lambda_2 \left[s\beta_I I + \frac{t\beta_W W}{M + W} \right] - \lambda_3 \left[(1 - s)\beta_I I + \frac{(1 - t)\beta_W W}{M + W} \right] \quad (28)$$

$$\lambda'_2 = -\frac{dH}{dI} = \beta_I \lambda_1 U - \lambda_2 (s\beta_I U - \alpha) - \lambda_3 [(1 - s)\beta_I U + \alpha] - \sigma \lambda_4 - c_0 \quad (29)$$

$$\lambda'_3 = -\frac{dH}{dD} = (\lambda_3 - \lambda_2)\gamma \quad (30)$$

$$\lambda'_4 = -\frac{dH}{dW} = \frac{\beta_W \lambda_1 U M}{(M + W)^2} - \frac{t\beta_W \lambda_2 U M}{(M + W)^2} - \frac{(1 - t)\beta_W \lambda_3 U M}{(M + W)^2} + \delta \lambda_4 \quad (31)$$

We characterize the optimal control by taking the derivative of the Hamiltonian with respect to the control, k . We have the following:

$$0 = \frac{dH}{dk} = -c_1 - 2c_2k + \lambda_4.$$

After solving for k , we get:

$$k^* = \frac{\lambda_4 - c_1}{2c_2}.$$

7.2 Simulations

7.3 Conclusions

8 Overall Conclusion

Our goal isn't to find the "best" one, just to look at different ones and see what the strengths and weaknesses of each are

8.1 How to add Citations and a References List

You can simply upload a `.bib` file containing your BibTeX entries, created with a tool such as JabRef. You can then cite entries from it, like this: [\[Gre93\]](#). Just remember to specify a bibliography style, as well as the filename of the `.bib`. You can find a [video tutorial here](#) to learn more about BibTeX.

If you have an [upgraded account](#), you can also import your Mendeley or Zotero library directly as a `.bib` file, via the upload menu in the file-tree.

References

- [Gre93] George D. Greenwade. The Comprehensive Tex Archive Network (CTAN). *TUGBoat*, 14(3):342–351, 1993.